METRIC

A-A-59003/4
31 May 1995
SUPERSEDING
MIL-A-21577/4
15 August 1968

COMMERCIAL ITEM DESCRIPTION

AMPLIFIER-OSCILLATOR GROUP AN/SIA-117

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-A-21577/4B which is canceled.

1. **Scope**. This commercial item description covers the detailed requirements for an audio amplifier-oscillator group AN/SIA-117 for general announcing systems, circuits 1MC, and aviators' announcing systems, circuit 3MC, for use on aircraft carriers.

2. **Salient Characteristics**

2.1 **General**. Amplifier-oscillator group AN/SIA-117 shall be in accordance with the applicable paragraphs of A-A-59003 and as specified herein. Whenever a requirement of A-A-59003 conflicts with a requirement of this specification, the requirement of this specification shall govern.

2.1.1 **Digital circuits**. Microphone preamplifiers, alarm oscillators, alarm output preamplifiers, VU meters, and any associated internal electronics shall utilize digital electronic technology.

2.1.2 **Speech processor**. A speech processor shall be incorporated into the audio circuit path. The speech processor circuit shall be a separate card from any other audio card. Insertion of or removal of the processor shall not disable normal 1MC or 6MC operation. The speech processor output shall be for engineering space loudspeaker groups only; all other loudspeaker output groups shall have unprocessed audio. The processor shall operate as specified in the acquisition specification.

2.2 **Equipment**. Equipment complement shall consist of the following:

(a) One common enclosure, bulkhead mounted, amplifier control rack, dimensions not to exceed 915 mm high, 610 mm wide, and 534 mm deep, containing:

2 Input preamplifier modules.
2 Alarm output preamplifier modules.
2 Power supply modules.
2 Oscillator modules (collision alarm).
2 Oscillator modules (chemical alarm).
2 Oscillator modules (general alarm).
4 Oscillator modules (unassigned).

AMSC N/A

FSC 6320

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All necessary switches, meters, relays, wiring, terminals, and other components required to control three AM-2316/SIA amplifier assemblies; two connected for normal 1MC-3MC operation and the third connected for an emergency alert system.

All components required to connect to a remote AN/SIA-119 amplifier-control group to provide for its use as an emergency amplifier.

2.3 **Construction**. Modular construction techniques with quick disconnect features shall be employed in the design of these assemblies to permit easy removal of the module for servicing.

2.4 **Module mounting**. Modules shall be mounted in a vertical position to facilitate the unrestricted flow of air between modules for ventilation purposes.

2.5 **Semiconductor devices**. Semiconductor devices shall be used exclusively.

2.6 **Forced air cooling**. Forced air cooling will not be permitted.

2.7 **Time constant**. Under the conditions specified in the overload limiter characteristic requirements of A-A-59003, the preamplifier output voltage shall return to the value specified in the overload limiter characteristic requirements within 0.025 second after the sudden increase in input signal voltage above reference level.

2.8 **Test facilities**. A common voltmeter with switching facilities shall be provided on the control rack to determine proper power supply voltage for amplifier-oscillators and preamplifiers. The meter calibration and circuit component values shall be such that the operating condition of the amplifiers or amplifier-oscillators can be determined directly from the meter without reference to tables or charts.

2.9 **Interstage input facilities**. Interstage input circuit, input impedance, and input level will not be required.

2.10 **Alarm signals**. The amplifier-oscillator modules shall generate the following alarm signals:

Signal number	Signal function
1	Collision alarm
4	Chemical alarm
5	General alarm
7	Unassigned
8	Unassigned

2.10.1 **Generated signals**. Alarm signals numbers 2, 3, and 6 will not be required. Electronic timing shall be provided on circuits where timing is required.

2.10.2 **Emergency alert**. Operation of an external alarm contact maker shall connect output of 1MC amplifier to all connected loudspeakers and output of emergency alert amplifier to emergency alert loudspeakers. Connection to all loudspeakers shall be maintained for subsequent voice announcements until system is manually reset to normal at the end of the emergency.

2.11 **Output level**. The amplifier-oscillator shall maintain an output

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failure alarm shall be capable of [illegible] The three phase bus

level of not less than 10 volts into a 600-ohm resistive load for amplifier input. This voltage shall also be used as a test signal source in accordance with A-A-59003.

2.12 **Adjustment of output level**. A lockable adjustment, adjustable by screwdriver, coin, or hand, shall be provided in the amplifier-oscillator or in combination with the associated control circuits to permit independent adjustment of the amplifier-oscillator output level for the alarm signal into the amplifier input.

2.13 **Input facilities**. Amplifier-oscillator group AN/SIA-117 shall have the following input facilities:

Microphone station number	Microphone station type	Circuits controlled
1 through 8	IC/SMB-2	1MC
9 through 15	IC/SMB-2	1MC

2.14 **Circuit attenuation**. Operation of the 1MC system for any microphone station shall energize a circuit that is capable of attenuating circuit SE (Ship's Entertainment system). This circuit shall be capable of making and breaking a resistive load of 7 amperes at 115 volts, 60 cycles, single phase.

2.15 **Local loudspeaker cutout circuit**. A cutout circuit shall be included for each microphone station to automatically disconnect local loudspeakers when the microphone is in use. The cutout circuit shall consist of a normally closed relay which shall operate to open the audio circuit supplying the local loudspeakers when the microphone press-to-talk switch is operated. These relays shall be installed in the amplifier control rack. Fifteen local cutout circuits are required.

2.16 **Output facilities**. Amplifier-oscillator group AN/SIA-117 shall have the following output facilities:

Loudspeaker Group No.	Loudspeaker Group Designation	Circuit
1	Officers	1MC
2	Topside	1MC
3	Crew	1MC
4	Engineering spaces	1MC
5	Flight deck walkways	1MC and 3MC
6	Manager deck walkways	1MC and 3MC
7	Emergency alert (Officers)	Emergency alert
8	Emergency alert (Crew)	Emergency alert

2.17 **Loudspeaker group connection**. The loudspeaker groups specified in 3.19 shall be connected to each 500-watt amplifier assembly where two amplifiers are utilized. Circuit 1MC shall have priority over circuit 3MC.

2.18 **Metering and switching**. Input and output metering and switching shall apply as specified in 2.19.

2.19 **Combined metering**. A common volume indicator meter shall be provided for input and output metering. This meter shall indicate zero decibels (db) at rated input and output power.

2.20 **Test input level control**. Test input level control will not be required.

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2.21 **Microphone press-to-talk switches**. Reduction in output of circuit SE following operation of a press-to-talk switch, shall not apply to the 3MC system.

2.22 **Visual alarm**. Operation of alarm signals numbers 1, 4, 5, 7, and 8 shall initiate operation of the visual alarm signal.

2.23 **Circuit SE muting**. Operation of any alarm contact shall energize a circuit that is capable of muting circuit SE. This circuit shall be capable of making and breaking a resistive load of 7 amperes at 115 volts, 60 cycles, single phase.

2.24 **Alarm signal priorities**. The order of priority for alarm signals shall be as follows:

(a) Collision alarm.
(b) Chemical attack alarm.
(c) General alarm.
(d) Unassigned.
(e) Unassigned.

The oscillator modules shall be provided with a device to prevent unauthorized or accidental changes in alarm signal priorities. This device shall allow intentional change in alarm priority by simple mechanical change.

3. **Quality assurance**. Sampling, examination, and tests shall be in accordance with A-A-59003.

4. **Packaging**. Packaging shall be in accordance with A-A-59003.

5. **Notes**

5.1 **Intended use**. Amplifier-oscillator group AN/SIA-117 is intended for use in shipboard general and aviators' announcing systems on aircraft carriers.

5.2 **Notes**. The notes specified in A-A-59003 are applicable to this specification.

Preparing activity:
Navy - SH
(Project 6320-N050)

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